

Database Connection Pool Exhaustion Runbook

Purpose

This runbook explains how to investigate and remediate database connection pool exhaustion in the ecommerce application.

Symptoms

- HTTP 500 errors from the application
- Increased request latency
- Database connection timeout errors
- Application logs containing "connection pool exhausted"
- Increasing number of waiting database connections

Investigation

- 1 Check application logs for database timeout and connection pool errors.
- 2 Check the number of active database connections.
- 3 Check whether a recent application deployment changed database connection settings.
- 4 Check for long-running database queries.
- 5 Check whether application instances are creating connections faster than the database can handle.

Remediation

- 1 Identify the application instances consuming excessive connections.
- 2 Check for leaked or long-lived database connections.
- 3 Review the application's connection pool configuration.
- 4 If an application instance is stuck and approved for restart, restart that instance using the approved operational procedure.
- 5 Verify that database connections return to normal levels.
- 6 Monitor application latency and error rates after remediation.

Verification

- Database connection utilization returns to normal.
- Application errors decrease.
- Request latency returns to normal.
- No new connection timeout errors are observed.

Prevention

- Configure appropriate connection pool limits.
- Set connection timeouts.
- Monitor active database connections.
- Monitor long-running queries.

- Include database connection metrics in production dashboards.
- Review connection-pool configuration during application deployments.

Example

If the ecommerce API suddenly starts returning HTTP 500 errors and logs show "database connection pool exhausted", the engineer should first verify active database connections and application logs before restarting services.

All remediation actions should be recorded in the incident ticket.